# AI Usage Declaration

**Project Title:** Predictive Analysis System for Karachi Air Quality

## AI Tool Used

- **ChatGPT (OpenAI)**

## Purpose of AI Usage

ChatGPT was used as a learning and debugging assistant during the project development. Its role was limited to reviewing the machine learning workflow, and helping identify a data leakage issue that affected the model's evaluation.

## Sections Assisted by AI

The following aspect of the project involved AI assistance:

- **Data Leakage Detection and Resolution:** ChatGPT helped identify that the engineered feature **`Pollution_Index = PM25 + PM1`** introduced data leakage because it directly contained the target variable (**PM2.5**). Based on this explanation, I removed the feature, retrained the models, and re-evaluated their performance to obtain realistic results.

## Work Completed Independently

The following work was completed by me without AI-generated implementation:

- Selection of the project topic and objectives.
- Collection and use of the Aga Khan University air quality dataset.
- Data loading, preprocessing, and cleaning.
- Creation of the pivot table and feature extraction (hour and day).
- Handling missing values.
- Feature engineering (`Temp_Humidity_Ratio`).
- Implementation of the machine learning pipeline.
- Training and evaluation of Linear Regression, Decision Tree Regressor, and Random Forest Regressor models.
- Comparison of model performance using $R^2$ Score and Mean Squared Error (MSE).
- Development of the interactive PM2.5 prediction interface using `ipywidgets`.

## Declaration

I confirm that this project was primarily designed, implemented, and tested by me. AI assistance was used only to improve my understanding of a specific machine learning issue (data leakage) and to support the debugging process. All final implementation, code modifications, model training, testing, and project outcomes are my own work.